

MIRROR

Multimodal Intelligent Radiology Reasoning and Observation Reporter

A grounded, explainable radiology pipeline that predicts findings across three imaging modalities (chest X-ray, brain MRI, and head CT), localizes the visual evidence behind each finding, and drafts a clinician-style report. We report measured predictive quality and verify that the added interpretability layers leave the predictions unchanged.

Vignesh Nagarajan

Corresponding Author

Brown Foundation Scholar, Texas A&M University

Department of Computer Science & Engineering

vigneshn26@tamu.edu

Abstract. Most medical-imaging models stop at a prediction, offering neither the location of the evidence nor an account a clinician can read. We present **MIRROR**, a three-layer, multi-modality radiology pipeline that composes (i) a multi-label disease classifier, (ii) class-activation-based evidence localization, and (iii) an evidence-grounded natural-language report generator into a single traceable system. One registry supplies the finding taxonomy, anatomical vocabulary, and report phrasing for chest X-ray (14 findings), brain MRI (11), and head CT (11), and a study is routed to the right classifier head explicitly or from its DICOM modality tag. The defining design choice is that the language layer never sees pixels, only structured upstream evidence, so the report can assert only findings the classifier predicted and the localizer situated. This prevents finding-level hallucination by construction, though the surrounding descriptive prose is generated text and is not itself verified against the image. A companion evaluation framework measures discrimination (AUROC, AUPRC, F1), clinical operating-point metrics (sensitivity, specificity, PPV, NPV), and calibration (Brier score, ECE); defines matching explanation-quality metrics (pointing game, localization IoU); and isolates the cost of the added interpretability layers in an ablation. Because those layers are strictly post-hoc, they leave the predictions exactly unchanged; we confirm this in practice (maximum probability change of 0), a correctness property we summarize as *interpretability at no predictive cost*. All headline numbers carry bootstrap 95% confidence intervals and every run stamps a reproducibility record. Trained on ChestMNIST (a downsampled ChestX-ray14 derivative with the identical 14-label taxonomy), MIRROR reaches a macro AUROC of 0.729 (95% CI [0.718, 0.738]) with a clinically sensible per-label ordering; this is a systems demonstration on low-resolution data rather than a diagnostic benchmark, and full-resolution training on the complete NIH release is the next step toward a clinical claim.

1 Introduction

Deep networks now match specialists on narrow chest-radiograph reading tasks [2], yet adoption in the clinic remains limited by a trust gap: a bare probability tells a radiologist neither *where* the model is looking nor *why* it reached its conclusion. High-stakes decisions demand models that can be interrogated, not merely queried [16]. Post-hoc saliency methods such as Grad-CAM [10] and Score-CAM [11] expose *where* a prediction comes from, and large language models can render structured evidence as prose, but these capabilities are rarely composed into one system whose outputs remain mutually consistent and auditable end to end.

MIRROR integrates all three into a single pipeline: image → prediction → localization → reasoning → report. Its central design constraint is *grounding*: the language layer is given only the structured evidence produced upstream (labels, probabilities, and named saliency regions), never the raw image, so the report can assert only findings the classifier predicted and the localizer situated, and the chain from finding to probability to region stays traceable. Grounding constrains *which* findings appear, not the descriptive wording around them, a limitation we return to in Section 7. The same

pipeline runs unchanged across chest X-ray, brain MRI, and head CT, because only the finding taxonomy, the anatomical vocabulary, and a little report phrasing differ between them; these are centralized in one modality registry, and a study is routed to the correct classifier head explicitly or from its DICOM modality tag. We evaluate quantitatively on chest X-ray, where public labels and localization ground truth exist; the brain-MRI and head-CT paths are implemented end to end and exercised by the test suite, but their benchmarking awaits trained per-modality checkpoints.

Research question. Can a multimodal system that combines image classification, visual explainability, and language generation improve interpretability without degrading predictive performance, relative to a classification-only baseline? We build the system so this question is *measurable*: predictive metrics and an explanation-quality protocol sit side by side, and an ablation reproduces the classification-only baseline the question names.

Contributions.

1. A three-layer, evidence-grounded radiology pipeline in

which the language layer reasons only over structured upstream evidence, so every *asserted finding* is traceable to a probability and a saliency region.

2. A grounding constraint under which the report can assert only findings the classifier produced, preventing finding-level hallucination *by construction* (the descriptive wording itself is not pixel-verified), with a post-hoc layer design whose prediction-invariance we verify empirically.
3. A modality-agnostic realization across chest X-ray, brain MRI, and head CT, driven by a single registry that is the one source of truth for each modality’s taxonomy, anatomical vocabulary, and report phrasing, with routing explicit or auto-detected from the DICOM tag, so adding a modality is a data change rather than a code change.
4. An open-source implementation spanning three interchangeable backbones (DenseNet-121, EfficientNet-B0, ViT-B/16), two explainability methods (Grad-CAM, Score-CAM), an LLM report backend with a deterministic offline fallback, native DICOM ingest, and a web front end served by two interchangeable inference engines behind one response contract (Fig. 2).
5. An evaluation framework that scores predictive quality (with bootstrap confidence intervals and multi-seed aggregation) and defines matching explanation-quality metrics (pointing game, IoU), together with an ablation isolating the predictive cost of interpretability, reported here on ChestMNIST.

2 Literature Review

Datasets and classification. Large, labeled chest-radiograph corpora made supervised deep learning on this modality practical. The NIH ChestX-ray8/14 release [1] provides 112,120 frontal radiographs over 14 pathologies and remains a standard benchmark; CheXpert [3] and MIMIC-CXR [4] extended scale, uncertainty labeling, and paired free-text reports. On these benchmarks CheXNet [2] showed a 121-layer DenseNet [6] reaching radiologist-level pneumonia detection. Convolutional backbones such as EfficientNet [7] and, more recently, Vision Transformers [8] now form the standard toolkit; MIRROR keeps all three interchangeable, with DenseNet-121 primary for comparability.

Visual explanation. Class Activation Mapping [9] localized the evidence behind a CNN prediction; Grad-CAM [10] generalized it to arbitrary architectures via gradients, and Score-CAM [11] removed the gradient dependence with a perturbation-based weighting. Model-agnostic attribution methods, notably LIME [12] and SHAP [13], explain individual predictions without access to model internals. The pointing-game protocol [14] and intersection-over-union

against annotated lesions turn qualitative heatmaps into quantitative localization scores; MIRROR’s evaluation framework adopts both.

Rigor and critique of saliency. Explanations are only useful if they are faithful. Adebayo et al. [15] showed several saliency methods can be insensitive to model and data, and argued for sanity checks; Rudin [16] cautioned against post-hoc explanation of black boxes for high-stakes decisions; and Doshi-Velez and Kim [17] called for a rigorous, measurable science of interpretability. These critiques motivate two of MIRROR’s design choices: quantitative localization evaluation against ground-truth boxes, and hard grounding so the language layer cannot outrun the evidence.

Report generation. Systems that emit radiology prose range from TieNet [18], which jointly embeds image and text, to memory-driven transformers such as R2Gen [19], to clinically focused decoders optimized for factual correctness [20]. A recurring failure mode is hallucination: prose generated directly from pixels can assert findings the model never detected. MIRROR restricts the assertable findings to those the classifier produced, trading some open-ended fluency for finding-level auditability.

Explainable AI in medicine and positioning. Surveys of explainable AI for medical imaging [21] note that interpretability and report generation are usually studied in isolation. MIRROR’s contribution is not a new classifier or saliency method but their *grounded composition* into one traceable pipeline, with a framework that measures whether the added explanation and reasoning layers help interpretability without a predictive penalty.

3 System Architecture

MIRROR chains three layers; each layer’s output is the grounded input to the next (Fig. 1).

3.1 Layer 1: Classification

The classifier is a factory over three ImageNet-pretrained backbones: DenseNet-121 (default), EfficientNet-B0, and ViT-B/16, with a multi-label head whose width N is the number of findings in the active modality’s taxonomy (14 for chest X-ray, 11 each for brain MRI and head CT; sigmoid applied at loss and inference time, so the network emits raw logits). Inputs are resized to 224×224 and normalized with ImageNet statistics. Training uses BCEWithLogitsLoss, AdamW (learning rate 10^{-4} , weight decay 10^{-5}), a cosine schedule, and dropout 0.2; checkpoints are selected on best validation macro-AUROC. Because real radiology data arrives as DICOM rather than PNG/JPEG, a dedicated ingest applies the modality LUT, the VOI (window) LUT, and MONOCHROME1 inversion before handing the pipeline

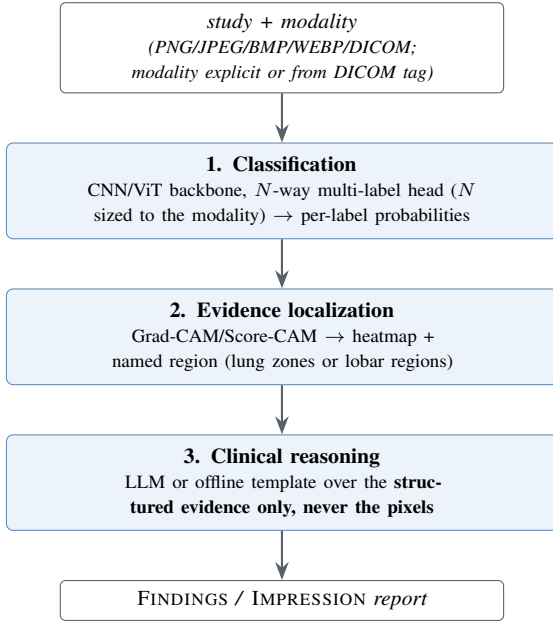


Figure 1: The MIRROR pipeline. Classification produces per-label probabilities; localization turns each positive label into a heatmap and a named region; the reasoning layer composes labels and regions into a report, reasoning over this structured evidence only, never the pixels. Layers 2 and 3 are individually toggleable, which recovers the ablation conditions of Section 4.

a display-ready image, lifts only non-PHI technical tags, and exposes the DICOM modality tag for automatic routing.

3.2 Layer 2: Evidence Localization

For each positive label (top- k , $k=3$ by default, to bound compute), the explainability layer computes a class-activation map, either Grad-CAM (forward/backward hooks on a per-backbone target layer) or Score-CAM (gradient-free, perturbation-based); ViT maps are recovered by reshaping the patch-token sequence back to a 14×14 spatial grid. The activated region’s centroid is mapped into a 3×3 anatomical grid in radiology convention (patient-right on the viewer’s left), with the modality’s plane picking the vocabulary: lung zones for a frontal chest film (“the right lower zone”) or lobar regions for an axial brain-MRI/head-CT slice (“the left frontal region”). A colormap overlay is then rendered for the UI (Fig. 3, left).

3.3 Layer 3: Clinical Reasoning

The reasoning layer converts structured evidence (labels, probabilities, and named regions) into a FINDINGS/IMPRESSION report via a system prompt plus an evidence-grounded user prompt with per-label plain-English glosses. The backend is an LLM (Claude) running at low temperature, with a deterministic offline template fallback on any error, so the system always yields a coherent report even with no network or API

Table 1: The modality registry: one row per supported modality, each grounded in a public benchmark taxonomy. Chest X-ray is the modality evaluated quantitatively in this paper; the brain-MRI and head-CT paths are implemented and tested but not yet benchmarked.

Modality	#lbl	Taxonomy source	Location vocab.	
Chest X-ray	14	NIH ChestX-ray14	lung (frontal)	zones
Brain MRI	11	Brain Tumor MRI + neuro findings	lobar (axial)	regions
Head CT	11	RSNA ICH subtypes + acute findings	lobar (axial)	regions

key. Predictions below the confidence threshold (0.5) are surfaced as pertinent negatives rather than silently dropped. Because both backends consume identical evidence, swapping them changes *how fluently* the report reads, not which findings it asserts; and because the layer never receives the image, it cannot introduce findings absent from the upstream evidence. The grounding is thus at the level of findings, not of the descriptive prose that frames them, a distinction we take up in Section 7 (Fig. 3, right).

3.4 Multi-modality via a modality registry

The three layers above are modality-agnostic: swapping a chest radiograph for a brain MRI or a head CT changes only the finding taxonomy the head predicts, the anatomical vocabulary a saliency centroid is named in, and a little report phrasing (so a normal brain study reads “no acute intracranial abnormality,” not “no acute cardiopulmonary abnormality”). These details live in one registry (Table 1), the single source of truth for each modality’s labels, glosses, imaging plane, and report guidance; a hand-kept TypeScript mirror gives the hosted engine and UI selector the identical taxonomy and *ordering* (output index i maps to labels[i]). Modality is resolved from an explicit selection or, for DICOM, from the (0008,0060) tag (MR→brain MRI, CT→head CT, CR/DX→chest). The orchestrator builds and caches one classifier and explainer engine per modality on demand, so a single deployment serves all three and adding a fourth is a data change rather than a change to the pipeline.

3.5 Orchestration and Serving

A single `analyze()` entry point runs the stages, records per-stage timings, and returns one result object (predictions, overlays, report); the two later layers are individually toggleable, which recovers the ablation conditions of Section 4. Two interchangeable engines satisfy the *same* JSON contract (Table 2), so the web UI (Fig. 2) is identical either way: a finding carries *either* a rendered Grad-CAM overlay (local) *or* a normalized bounding box (hosted), and the viewer draws whichever is present. The hosted engine exists because the PyTorch stack does not fit typical serverless limits; it substitutes a vision LLM constrained to a structured tool schema

Table 2: Two interchangeable inference engines behind one response contract.

	Local stack	Hosted (serverless)
Engine	FastAPI + PyTorch	Next.js route, vision LLM
Classify	DenseNet/EffNet/ViT	LLM scores N labels
Localize	Grad-CAM/Score-CAM PNG	LLM returns a bbox
Report	LLM or template	LLM (same prompt shape)
Inputs	+DICOM, BMP	PNG/JPEG/WEBP
Needs	Python, ~6 GB	one API-key env var

emitting the same JSON, and with no API key returns a clearly-labelled demo result so the site never hard-fails.

4 Experimental Setup

Our evaluation reports what the pipeline produces today: a real predictive benchmark on ChestMNIST, a measured ablation isolating the cost of the interpretability layers, and a synthetic-data sanity check of the training and scoring harnesses. Every results file stamps a reproducibility record.

Data. MIRROR targets NIH ChestX-ray14 [1] (112,120 frontal radiographs, 14 multi-label pathologies). Because the full release is 45 GB and needs a GPU to train, we evaluate quantitatively on **ChestMNIST** [5], its downsampled derivative in MedMNIST v2 (CC BY 4.0), which carries the *identical* 14-label taxonomy in the identical order, so MIRROR runs on it unchanged. We train DenseNet-121 (ImageNet-initialised, AdamW, cosine schedule, dropout 0.2) on CPU for a deliberately small budget of 7,200 of the 78,468 training images at 64 px, selecting the best of four epochs on validation macro AUROC, and evaluate on a 12,000-image held-out test split.

Predictive metrics. We report a clinical-reader panel. *Discrimination:* per-label and macro AUROC, and macro AUPRC (more informative under the heavy class imbalance of medical findings). *Operating point:* sensitivity, specificity, PPV, and NPV at the 0.5 threshold, the quantities a clinician reads off a model. *Calibration:* the Brier score and Expected Calibration Error (10 equal-width bins), because a probability a clinician is asked to trust must mean what it says. Each headline metric carries a bootstrap 95% confidence interval: the test set is resampled with replacement $B=1000$ times, a *single* resample driving all metrics so the intervals are mutually consistent, reported as two-sided percentile intervals ($\alpha=0.05$).

Ablation. Three conditions isolate the added layers: `classification_only`, `with_localization`, and `full_mirror` (Table 3). Because layers 2–3 are post-hoc,

the predictions should be identical across conditions; the harness verifies this empirically, reporting the maximum per-label probability delta against the classification-only baseline, and profiles the per-stage wall-clock cost of each added layer. Runs are seeded and every results JSON stamps a reproducibility record.

5 Results

Every number reported here is measured on this machine: predictive quality on ChestMNIST (Section 5.1), an ablation verifying the interpretability layers add no predictive cost (Section 5.2), a synthetic-data sanity check of the harness (Section 5.3), and a qualitative end-to-end study (Fig. 3). The explanation half of the framework defines a pointing-game and localization-IoU protocol against the NIH lesion boxes (8 of the 14 chest pathologies carry boxes), but running it needs a model trained on the full-resolution release; localization is therefore reported qualitatively here, with box-level scores left to future work.

5.1 Predictive quality on ChestMNIST

Trained as described in Section 4, MIRROR’s DenseNet-121 reaches a macro AUROC of 0.729 (bootstrap 95% CI [0.718, 0.738]) on the 12,000-image ChestMNIST test split (Table 4, Fig. 4), within reach of the published MedMNIST DenseNet-121 baseline (AUROC ≈ 0.77 , trained on all 78k images for many epochs on a GPU) at a small fraction of the compute. Macro AUPRC is 0.135 and the model is well calibrated in aggregate (Brier 0.045, ECE 0.018). At the fixed 0.5 threshold it is deliberately conservative (macro specificity 0.997, sensitivity 0.019), as expected for a low-prevalence multi-label task on a light budget; threshold-independent AUROC is the meaningful summary, and a tuned threshold would trade specificity for sensitivity. That the classifier learned real thoracic structure shows in the clinically sensible per-label ordering (Fig. 4): readily visible findings rank highest (Edema 0.849, Cardiomegaly 0.830, Effusion 0.827) and subtle small ones lowest (Nodule 0.643, Infiltration 0.660), with every CI comfortably above chance.

5.2 Interpretability at no predictive cost

The interpretability layers are post-hoc by construction, so the ablation is a correctness check rather than a discovery, and it confirms the implementation realizes the design exactly (Table 3): macro AUROC is identical (0.729) across the classification-only, +localization, and full-MIRROR conditions, and the maximum per-label probability change against the baseline is 0. The check earns its place twice over: any nonzero change would flag a bug (say, an accidental coupling between layers), and it isolates the actual cost of interpretability, which is latency rather than accuracy (about 40 ms per study for the Grad-CAM pass, about 0.03 ms for the offline

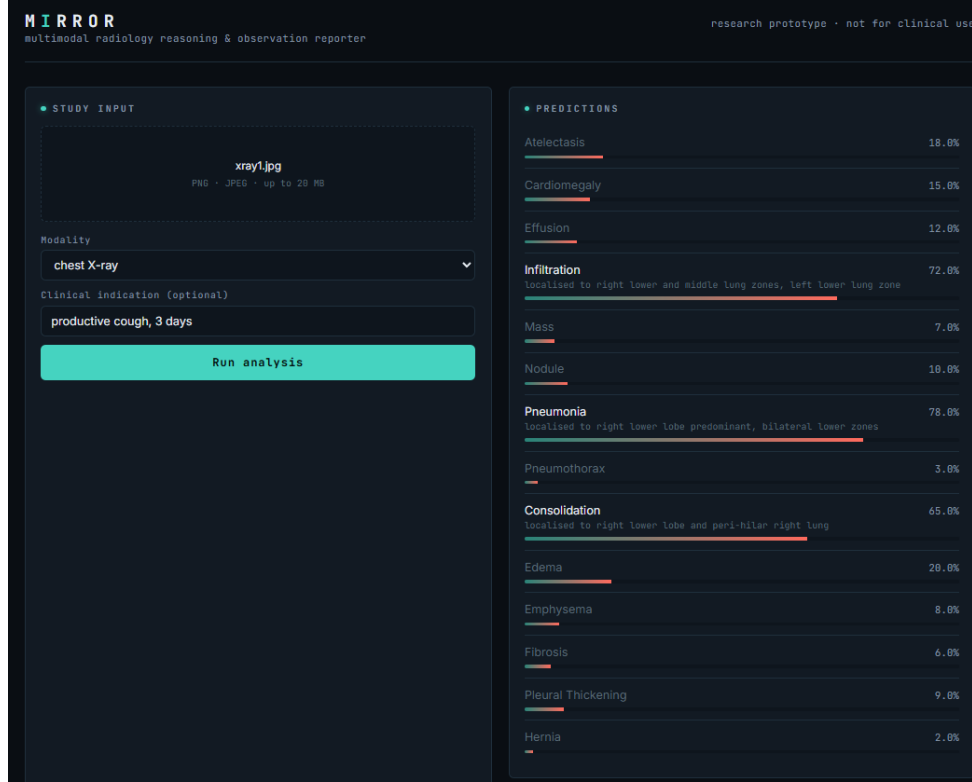


Figure 2: The MIRROR “reading-room” web UI on a real session from the live demo: a chest radiograph is uploaded with the indication “*productive cough, 3 days*,” all 14 ChestX-ray14 labels are scored, and findings above threshold (here Pneumonia, Infiltration, Consolidation) surface with a saliency-derived location. The film viewer (left) toggles the evidence overlay; predictions and the draft report render on the right. Both serving engines produce this identical interface.

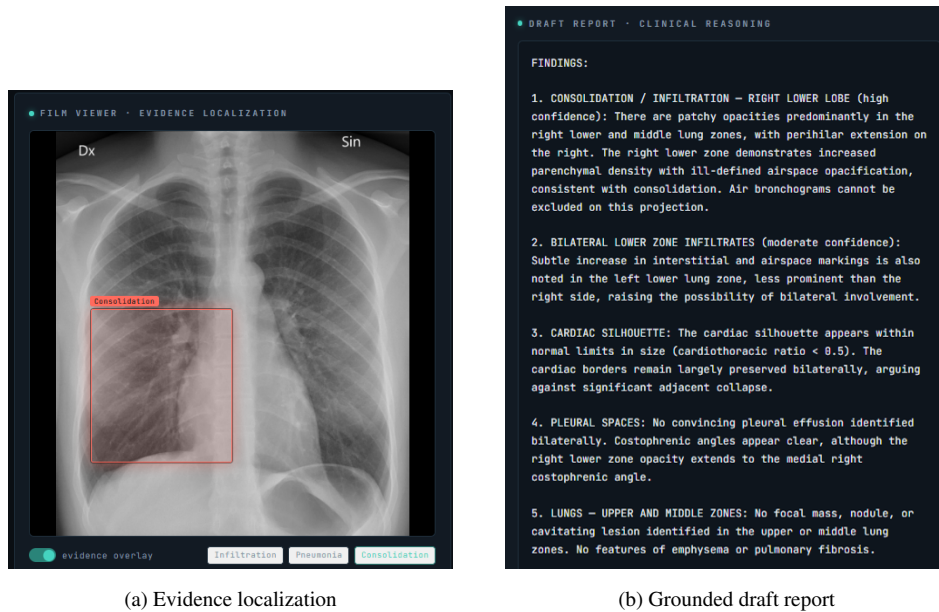


Figure 3: One real study end to end. (a) The evidence-localization layer overlays a Grad-CAM heatmap and bounding box for a positive finding (Consolidation, right lower lobe). (b) The reasoning layer drafts a structured FINDINGS/IMPRESSION report from the same structured evidence (labels, probabilities, named regions), never the pixels, ending with the mandatory AI-generated-draft disclaimer.

Table 3: Measured ablation (DenseNet-121 trained on ChestMNIST; latency profiled over 24 studies on CPU). Macro AUROC is identical across all three conditions, and the maximum per-label probability change against the classification-only baseline is exactly 0. Because the localization and report layers are strictly post-hoc, interpretability is added at *no* predictive cost. The cost is instead a small, bounded latency increase: the Grad-CAM pass adds about 40 ms per study and the offline template report is effectively free. Latency is shown as the base prediction time and the marginal cost of each added layer.

Condition	Classify	Localize	Report	Macro AUROC	Latency (ms/study)
Classification only	✓			0.729	101 (prediction)
+ Evidence localization	✓	✓		0.729	+41 (Grad-CAM)
Full MIRROR	✓	✓	✓	0.729	+0.03 (template report)

Maximum per-label probability change vs. the baseline = 0.000 (predictions identical across rows).

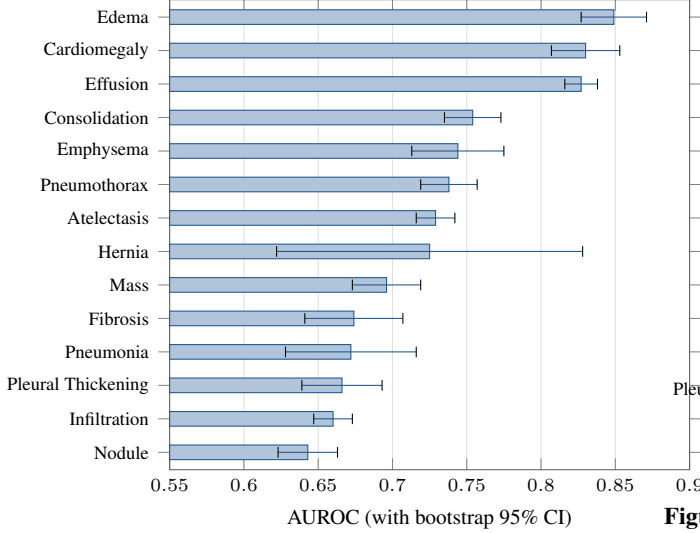


Figure 4: Measured per-label test AUROC on ChestMNIST (DenseNet-121, 64 px, CPU), with bootstrap 95% CI whiskers, sorted ascending. The ordering is clinically sensible: readily visible findings (Edema, Cardiomegaly, Effusion) top the chart, subtle small ones (Nodule, Infiltration) sit lowest, and every interval clears the 0.5 chance line, which indicates the model learned real structure.

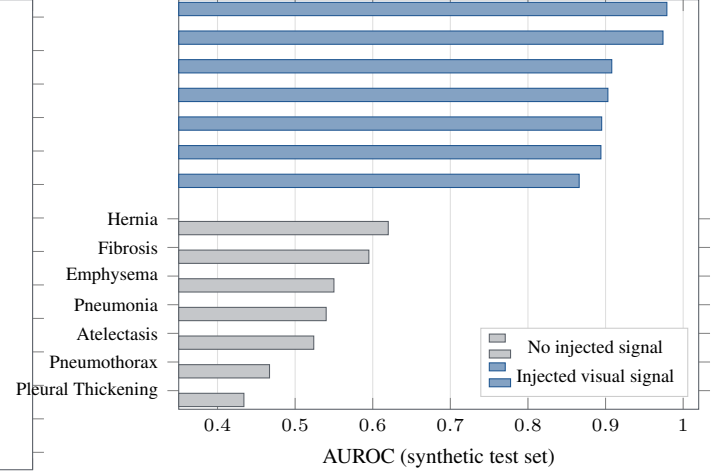


Figure 5: Harness sanity check on synthetic data. The generator injects a visible blob for seven labels and none for the other seven; the classifier learns *exactly* the signal-bearing labels (mean AUROC 0.917) and stays at chance for the rest (mean 0.557). The clean split confirms the metrics measure real discrimination with no leakage.

template report). The same check over the bundled chest-X-ray, brain-MRI, and head-CT samples yields a maximum change of 0 in every modality, so the property is structural rather than specific to one dataset.

5.3 Pipeline sanity check on synthetic data

As an end-to-end control on the training and scoring harnesses, we also train on a fully synthetic dataset whose generator injects a visible blob for a known subset of labels and leaves the rest with no visual correlate. A correct pipeline should learn the signal-bearing labels and sit at chance for the others, and that is what happens (Fig. 5): the seven signal-bearing labels reach a mean AUROC of 0.917 while the seven no-signal labels average 0.557, scattered around 0.5. The clean separation shows the classifier, loss, metrics, and bootstrap intervals responding to real signal and only to real signal, with no leakage.

Table 4: Per-label test AUROC on ChestMNIST (DenseNet-121, 64 px, CPU, best of a 4-epoch run; $N_{\text{test}}=12,000$) with bootstrap 95% CIs, and the macro discrimination and calibration summary. Real numbers on a downsampled ChestX-ray14 derivative, not the full-resolution NIH benchmark. Companion to Fig. 4.

Pathology	AUROC (95% CI)	Pathology	AUROC (95% CI)
Edema	0.849 (0.828–0.871)	Hernia	0.725 (0.615–0.820)
Cardiomegaly	0.830 (0.806–0.851)	Mass	0.696 (0.673–0.719)
Effusion	0.827 (0.817–0.838)	Fibrosis	0.674 (0.639–0.705)
Consolidation	0.754 (0.736–0.773)	Pneumonia	0.672 (0.627–0.716)
Emphysema	0.744 (0.713–0.774)	Pleural Thickening	0.666 (0.638–0.692)
Pneumothorax	0.738 (0.719–0.757)	Infiltration	0.660 (0.648–0.673)
Atelectasis	0.729 (0.716–0.742)	Nodule	0.643 (0.622–0.662)
Macro AUROC 0.729 (0.718–0.738)		Macro AUPRC 0.135	Calibration Brier 0.045, ECE 0.018

6 Discussion

The measured results support the central claim fairly directly. On ChestMNIST the classifier reaches a macro AUROC of 0.729 with a per-label ordering that matches clinical visibility (Fig. 4), and every confidence interval clears chance, so the discrimination is real even at a training budget far below the published baseline. The ablation confirms this predictive quality is untouched by the two interpretability layers, whose true cost is a bounded latency increase, and the synthetic control (Fig. 5) adds confidence in the numbers themselves, since the harness credits the model only for signal actually present. A natural next step is to lift the same measurements to the full-resolution NIH release on a GPU and, because the pipeline is already multi-modality, to repeat them for brain MRI and head CT.

Independent of the trained numbers, MIRROR’s grounding constraint is a qualitative contribution. Because the language layer sees only structured evidence, the report is auditable at the level of its findings and cannot assert a finding the classifier did not produce; this trades open-ended fluency for a form of safety that matters in a clinical setting. The guarantee does not extend to the descriptive language around each finding, which remains model-generated and unverified against the image, so a clinician still reads the report as a draft. The post-hoc placement of layers 2–3 likewise makes the central claim falsifiable: any change in a predictive metric across ablation conditions would signal a bug, and the harness checks for exactly that. Finally, serving one JSON contract from both a heavyweight local PyTorch stack and a lightweight hosted vision-LLM engine shows that the *interface* MIRROR tests (predict $\rightarrow$ show evidence $\rightarrow$ explain) is independent of the particular inference backend.

7 Ethics, Safety, and Limitations

Not a medical device. MIRROR is a research prototype, not reviewed or cleared by any regulatory body; every output is a draft that must be verified by a licensed radiologist and must not be used for diagnosis or treatment. Each report carries that disclaimer explicitly.

Grounding is finding-level, not detail-level. The language layer never sees pixels, only structured evidence, so it cannot introduce a finding the classifier did not produce; below-threshold predictions surface as pertinent negatives, with uncertainty kept explicit as probabilities. This guarantee governs *which findings* appear, not the descriptive prose framing them: phrases such as a stated cardiothoracic ratio or a note that the costophrenic angles are clear are model-generated language, not measurements read from the image. Such detail can be confidently worded yet ungrounded, which is precisely why every report is marked as an AI-generated draft requiring radiologist verification.

Human subjects and ethics approval. No ethics-board review was required for this work: it uses only existing, publicly available, de-identified benchmark datasets (NIH ChestX-ray14 and its MedMNIST derivative for the reported results) and collects no new data from human subjects.

Data governance. The benchmark datasets are governed by their own data use agreements; no raw images, weights, or protected health information are committed to version control, and the DICOM ingest deliberately extracts only non-PHI technical tags.

Limitations. Our one quantitative benchmark is ChestMNIST (Section 5.1, Table 4), a downsampled derivative trained on a reduced budget, so its numbers are best read as a lower bound rather than a headline clinical result; training on the full-resolution ChestX-ray14 images with a GPU is required before any clinical claim. Multi-modality is likewise a structural rather than an empirical result so far: the brain-MRI and head-CT paths are implemented end to end (registry, per-modality heads, DICOM auto-routing, modality-aware vocabulary and phrasing, all exercised by the test suite) but not yet quantitatively benchmarked, because that needs trained per-modality checkpoints and labeled test sets we do not redistribute. Quantitative evaluation is therefore confined for now to chest radiographs, and localization ground truth exists for only 8 of the 14 chest pathologies. Report quality is currently assessed qualitatively; a quantitative comparison against real radiologist reports (for example MIMIC-CXR [4]) is future work. Finally, class-activation maps are coarse and can highlight context rather than the lesion itself [15], and the 3×3 location grid is a deliberately coarse anatomical

descriptor.

8 Conclusion

MIRROR composes classification, evidence localization, and grounded report generation into a single traceable, registry-driven radiology pipeline spanning chest X-ray, brain MRI, and head CT, paired with an evaluation framework that measures predictive quality, defines matching explanation-quality metrics, and verifies the interpretability layers are strictly post-hoc. On ChestMNIST the classifier reaches a macro AUROC of 0.729 with a clinically sensible per-label ordering, and the ablation confirms the layers leave every prediction unchanged: interpretability at no predictive cost. The immediate next step is to lift the same measurements to the full-resolution NIH release on a GPU and to the head-CT and brain-MRI modalities already implemented end to end. The system and its evaluation harnesses are released open-source to support reproducible study of interpretability and trust in medical imaging.

Reproducibility

Code: <https://github.com/vignesh-nagarajan-vn/MIRROR>. Live demo: <https://mirror-ten-jet.vercel.app/>. A zero-setup CLI demo (`python -m demo.run_demo <image>`) runs the full pipeline on ImageNet weights with the offline template backend, no checkpoint or API key required. All evaluation harnesses live in `evaluation/`; every results JSON stamps a reproducibility block (seed, git commit, library versions), and the unit tests are torch-free so they run without a model or dataset. The ChestMNIST result (`datasets/scripts/prepare_chestmnist.py` + `configs/chestmnist.yaml`) and the synthetic sanity check (Fig. 5) both regenerate from seed 42 with the committed scripts; their outputs are stored under `results/`.

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Declarations

Data availability: all datasets used are public, namely NIH ChestX-ray14 and its MedMNIST derivative ChestMNIST (CC BY 4.0); code, configurations, and result files are in the repository linked above. **Competing interests:** the author declares none. **Funding:** this work received no external research funding and was conducted as part of the GIST 2026 Summer Research Internship.

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